E-Health for All:

A User-Centered Telemedicine System

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Alinday, Mark Joseph M.
Bruzo, Ian
Narvato, Dominnica SL.
Paraiso, Kristine Carla I.
Villamer, EJ Andrei J.

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College of Computer Studies
City of Naga

APPROVAL SHEET

This thesis project of **ALINDAY, MARK JOSEPH M.**, **BRUZO, IAN**, **NARVATO, DOMINNICA SL.**, **PARAISO, KRISTINE CARLA I.**, and **VILLAMER, EJ ANDREI J.** entitled **"E-HEALTH FOR ALL: A USER-CENTERED TELEMEDICINE SYSTEM"** was prepared and submitted in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Systems. It has been examined and is recommended for acceptance and approval for oral examination.

[ADVISER NAME]
Adviser

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[CHAIRMAN NAME]
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[PANELIST 2 NAME]
Panel Member

[SECRETARY NAME]
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Approved by the Committee on Oral Examinations on [DATE].

[CHAIRMAN NAME]
Chairman

[PANELIST 1 NAME]
Panel Member

[PANELIST 2 NAME]
Panel Member

Accepted and approved in partial fulfillment of the requirements for the degree of Bachelor of Science in Information Systems.

ANNA LORETTA C. ROMULO, PhD
Dean

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CHAPTER 1: INTRODUCTION

In the Philippines, these challenges are especially severe. The country's geography and urban concentration of medical specialists leave rural populations with limited health-care options. In response, the Philippine government and health institutions have begun embracing telemedicine to broaden coverage. The National Telehealth Center has launched SMS and web-based platforms connecting rural patients with specialist consultations [1], [2], and various health institutions have adopted electronic medical records to improve continuity of care across facilities [3], [4].

Despite these efforts, significant barriers persist. Poor transportation infrastructure hinders facility access, particularly during emergencies and follow-up care. Unreliable internet connectivity, limited computing resources, and low digital literacy among both patients and healthcare workers further compound these challenges [5], [6]. Existing telemedicine initiatives, while promising, remain fragmented and rarely address the unique technological constraints and cultural context of rural Filipino communities [7], [8].

This research seeks to bridge these gaps by developing an integrated telemedicine platform specifically designed for the selected rural barangays. The platform combines three components, a comprehensive doctors' database, a patient records management system, and an appointment scheduling system—into one user-friendly interface accessible to users with different levels of digital literacy.

This initiative aligns with multiple UN Sustainable Development Goals: Good Health and Well-Being (SDG 3), by improving healthcare access and quality; Industry, Innovation, and Infrastructure (SDG 9), by demonstrating how technology can address healthcare infrastructure gaps in underserved areas; and Reduced Inequalities (SDG 10), by examining how digital health solutions reduce disparities in healthcare access across different populations and socioeconomic groups.

1.1 Review of Related Literature

1.1.1 Telemedicine Adoption and Implementation

Telemedicine, the delivery of healthcare services using information and communication technologies, has emerged as a vital solution for healthcare access issues in isolated and underserved communities. Telemedicine allows healthcare providers to assess, diagnose, and treat patients in remote areas by utilizing telecommunications technology to effectively connect medical experts with populations that have limited access to healthcare facilities. It has shown great promise in overcoming healthcare access issues in rural communities around the world [9], [10].

Research shows that telemedicine consultations promote better clinical decision-making, reduce unnecessary patient transfers, and increase the chances of local admissions instead of costly transfers to far-off hospitals [11], [12]. These advantages apply to many medical fields, from primary care to specialized areas like neurology and emergency medicine [13].

However, the pandemic highlighted the shortcomings of telemedicine: access services at lower rates [14], [15]. Digital literacy among older adults and low-income individuals [5], [6], equal access concerns [16], [17], [18], implementation costs and maintenance [19], [20], infrastructure issues [21], [22], [23], and data security and patient privacy concerns [24].

On the one hand, communities with integrated systems achieve better health outcomes for managing chronic diseases compared to those that use fragmented record-keeping [25]. Patient satisfaction with telemedicine in the Philippines shows encouraging trends [25], [26]. Even senior citizens increasingly accepted remote consultations [27]. The University of the Philippines Health Service successfully used tailored electronic medical records systems in various settings during COVID-19, ensuring continuity of patient care despite mobility restrictions [2], [3]. Thorough assessments show considerable progress in adoption over time, although significant gaps between different types of facilities and locations persist [28], [29].

Successfully implementing telemedicine requires careful adjustments to local circumstances [30], [31]. Detailed analyses reveal Philippines-specific barriers like governance issues, unclear licensure requirements, unresolved liability problems, and vague reimbursement mechanisms that hinder healthcare provider involvement [1], [8]. Healthcare technology must consider local cultural practices, language preferences, and health-seeking behaviors to ensure meaningful user acceptance in diverse Filipino communities [7]. Creating user-friendly systems is crucial for sustained adoption rather than leaving behind abandoned pilot projects [32].

1.1.2 Patient Information Management System

Patient information management involves the organized collection, storage, and use of health data to support clinical decision-making and ensure continuity of care. Electronic Medical Records (EMR) systems digitize patient health information, enabling providers to access complete histories, track treatments, identify medication interactions, and coordinate care across specialties. This is especially critical in rural and underserved areas where access to prior care records is limited.

EMR systems significantly improve care coordination, with integrated communities achieving better chronic disease outcomes compared to those with fragmented records [33], [34], [35], [36]. However, while basic adoption rates are high, comprehensive implementation remains lower in rural facilities despite recognized benefits [17], [18]. Technical, organizational, and workflow barriers continue to hinder smooth information sharing [22], [28], [29], and digital literacy gaps among older adults and low-income individuals require targeted design and training interventions [5].

Data security and patient privacy are foundational to EMR trust, requiring encryption, access controls, and audit trails [24]. Integrated systems reduce administrative burdens and improve care delivery [37], [38], [39], while predictive models using patient data enable targeted interventions and better resource allocation [40], [41], supporting a shift toward proactive, preventive primary care [42].

In the Philippines, EMR adoption shows both progress and challenges. COVID-19 visibly impacted system usage in rural health facilities [43], yet implementation improved clinical decision-making and reduced medication errors compared to paper systems [4]. Sustained adoption is linked to training, technical support, and gradual workflow integration rather than rapid rollouts [44], though research representation across specialties, settings, and facility types remains uneven [45]. The absence of centralized provider databases causes referral delays and disjointed care [46], while comprehensive physician databases with specialization and scheduling details enhance diagnostic workflows [47]. Keeping such databases accurate remains challenging, necessitating automated verification and validation processes [48].

1.1.3 Appointment Scheduling in Healthcare

Appointment scheduling systems are essential to healthcare delivery, coordinating patient needs, provider availability, and facility resources. Well-designed systems reduce waiting times, lower missed appointments through automated reminders, improve provider productivity, and ensure urgent cases receive prompt attention. In rural settings, where patients face long travel distances and transportation barriers, efficient scheduling is especially critical for maintaining access and treatment adherence.

Digital scheduling significantly reduces no-show rates compared to offline booking, with automated reminders proving particularly effective [37], [49], [50]. AI-driven tools further improve efficiency by predicting non-attendance and optimizing resource allocation [38], while open-access scheduling reduces no-shows by offering same-day or next-day appointments [39]. Predictive models incorporating patient demographics and appointment history identify high lead time and prior missed appointments as the strongest predictors of non-attendance [40], [41]. Integrated systems also improve patient flow and reduce waiting area congestion in underserved facilities [51], supporting broader proactive primary care initiatives through timely referrals and fewer missed appointments [42], [48].

In the Philippines, digital appointment management in rural health centers has shortened waiting times and reduced missed appointments through better time slot management and automated reminders [52]. However, rural contexts require flexible approaches, as rigid scheduling is ineffective where patients face unreliable transportation or inflexible work schedules. Systems must therefore incorporate easy rescheduling options and reminder methods that function without internet access [53].

1.2 Synthesis of the State-of-the-Art

The detailed collection of works and studies offered valuable insights into the topic. These studies backed their claims with relevant data and detailed information.

Telemedicine has emerged as a viable solution to geographic barriers, improving clinical decision-making, reducing unnecessary transfers, and expanding specialist access [11], [12], [13]. However, persistent challenges remain, including infrastructure limitations, digital literacy gaps, data security concerns, and Philippines-specific barriers such as unclear licensure, liability, and reimbursement mechanisms [1], [6], [8], [24]. Despite these, large-scale pilots in the Philippines demonstrate feasibility across diverse communities, with patient satisfaction showing encouraging trends even among senior citizens [2], [25], [26], [27]. Successful adoption requires culturally sensitive, user-centered design tailored to local contexts [7], [32].

Patient information management systems significantly improve care coordination and clinical outcomes, yet comprehensive EMR implementation remains lower in rural facilities [17], [18]. Workflow barriers, digital literacy gaps, and data security concerns continue to hinder full adoption [5], [24], [28], [29]. In the Philippines, gradual integration supported by training and technical assistance yields better sustained adoption [44], while the absence of centralized provider databases causes referral delays and fragmented care [46], [47].

Appointment scheduling systems directly reduce no-show rates, improve patient flow, and support proactive primary care [37], [38], [42]. Predictive modeling and AI-driven tools further optimize resource allocation [38], [40], [41]. In Philippine rural health centers, digital scheduling has measurably shortened waiting times [52], though flexible, offline-capable systems are essential given transportation and connectivity constraints [53].

Across all three areas, the literature consistently points to the need for an integrated, context-sensitive platform that unifies patient records, provider information, and appointment scheduling—precisely the gap this research addresses.

1.3 Gap Bridged by the Study

This study addresses important gaps in rural healthcare delivery by creating an integrated telemedicine platform specifically for rural barangays in Naga City [54]. Existing telemedicine systems often look at patient management and appointment scheduling separately. This research offers a more complete solution by combining a doctors' database, a patient records management system, and appointment scheduling into one platform. It addresses the unreliable internet connectivity that is common in rural areas. Healthcare workers can keep entering data and accessing patient information even without internet access. Once the connection is restored, the system syncs automatically. By bringing these features together in one practical system designed for resource-limited settings, this research shows how telemedicine solutions can be adapted to meet the real technological, infrastructure, and cultural challenges faced in rural healthcare.

1.4 Theoretical Framework

This study is grounded in three key theories that collectively support the development of the proposed telemedicine system: the Diffusion of Innovations (DOI) Theory [55], the Task–Technology Fit (TTF) Theory [56], and the Technology Acceptance Model (TAM) [57]. Together, these theoretical foundations inform the system's design, usability considerations, and adoption potential within rural healthcare settings.

According to Goodhue and Thompson (1995), the Task Technology Fit (TTF) Model posits that technology will be adopted and effectively utilized when its functionalities align with the specific tasks that users are expected to perform. The model argues that a strong "fit" between task requirements—such as information retrieval, decision-making, communication, and documentation—and the capabilities of the system results in improved individual performance and greater likelihood of continued system use. When technology supports the user's workflow and reduces the effort required to complete essential activities, it becomes more valuable, usable, and integrated into daily operations. This model is relevant to the proposed telemedicine platform because its success depends on how well its features—such as patient record management, scheduling, and remote consultation support—match the operational needs and constraints of rural healthcare providers [56].

According to Rogers (2020), the Diffusion of Innovations (DOI) Theory explains how new ideas, practices, or technologies spread through a population over time. It emphasizes understanding the characteristics of innovations and the social processes that influence adoption rates among different groups [55]. The theory focuses on five key attributes that determine adoption: relative advantage, compatibility, complexity, trialability, and observability. In this study, it will guide the implementation strategy for the telemedicine platform in rural communities. The strategy will consider social dynamics and adoption patterns when introducing the system. The researcher can identify early adopters, such as progressive healthcare providers and influential community health workers, show its benefits to others, and help normalize its use in the community.

According to Marikyan and Papagiannidis (2024), the Technology Acceptance Model (TAM) asserts two significant factors in a user's intention to adopt a technology: perceived usefulness and perceived ease of use [57]. Perceived usefulness refers to the extent to which an individual believes that a technology enhances task performance and overall productivity, while perceived ease of use pertains to the belief that the system can be operated with minimal physical and cognitive effort. TAM is highly relevant to the proposed telemedicine platform because its adoption depends on users—especially those in rural areas with varying levels of digital literacy—perceiving the system as beneficial, simple to navigate, and supportive of their essential healthcare tasks.

1.5 Conceptual Framework

The enhanced system paradigm is the approach utilized in the conceptualization of this study. The input came from various sources to gain a complete understanding of rural healthcare needs and readiness for technology, such as document analysis of health records, observations of healthcare-seeking behaviors and travel times, surveys evaluating user satisfaction and technology skills, and interviews gathering in-depth information about access challenges and provider readiness for telemedicine services.

The processes involved include: systems analysis and specification for system design, functional testing to ensure adherence to system requirements, usability testing to ensure the system meets user needs, and acceptability validation to confirm system feasibility for deployment. The study's output is a web application with features including computer vision, security, authentication, verification, patient check-up history, and QR code security. The continuous feedback mechanism will ensure iterative refinement of the system to remain user-friendly, operational, and aligned with rural healthcare needs.

1.6 Statement of the Problem

This study aims to address the problem in healthcare accessibility of rural populations. Specifically, this study seeks to answer the following questions:

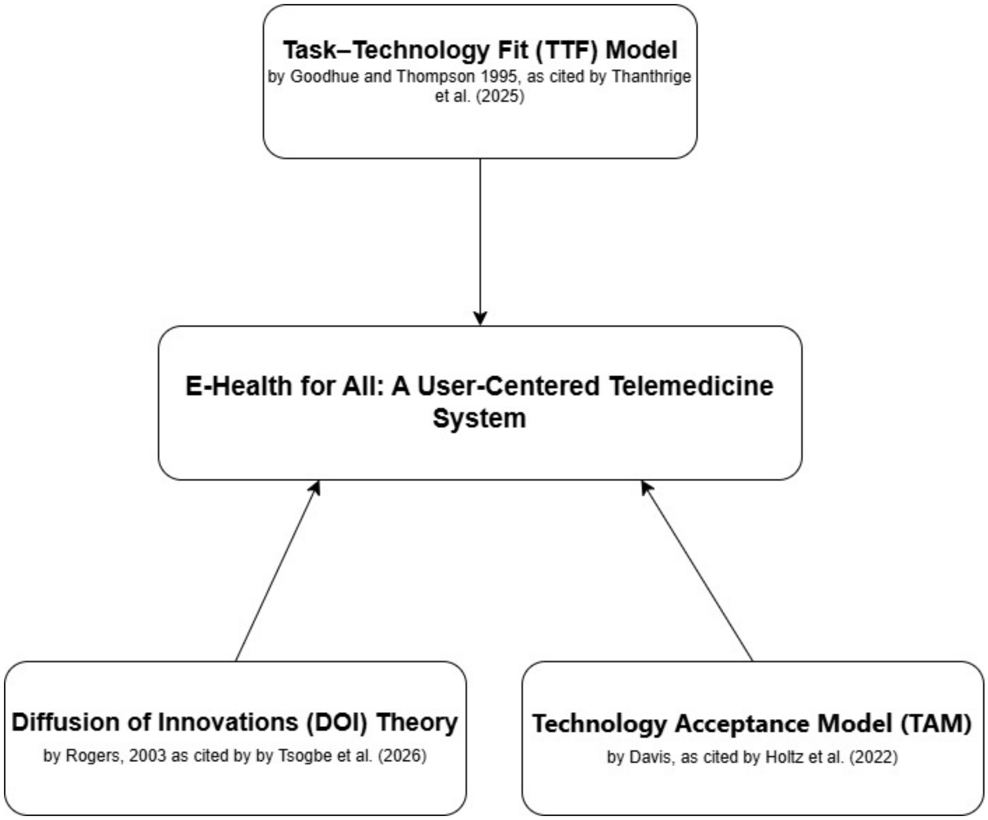

Figure 1.1. Theoretical Paradigm illustrating the DOI, TTF, and TAM frameworks guiding the E-Health for All telemedicine system development.

1. What are the current practices of rural healthcare providers in rural populations with respect to:
    a) Patient records management
    b) Doctor scheduling
    c) Appointment systems
2. What are the challenges encountered in the current system with respect to:
    a) Patient records management
    b) Doctor scheduling
    c) Appointment systems
3. What system can be designed to address the challenges encountered?
4. What is the level of acceptability of the system in terms of:
    a) Usability
    b) Reliability

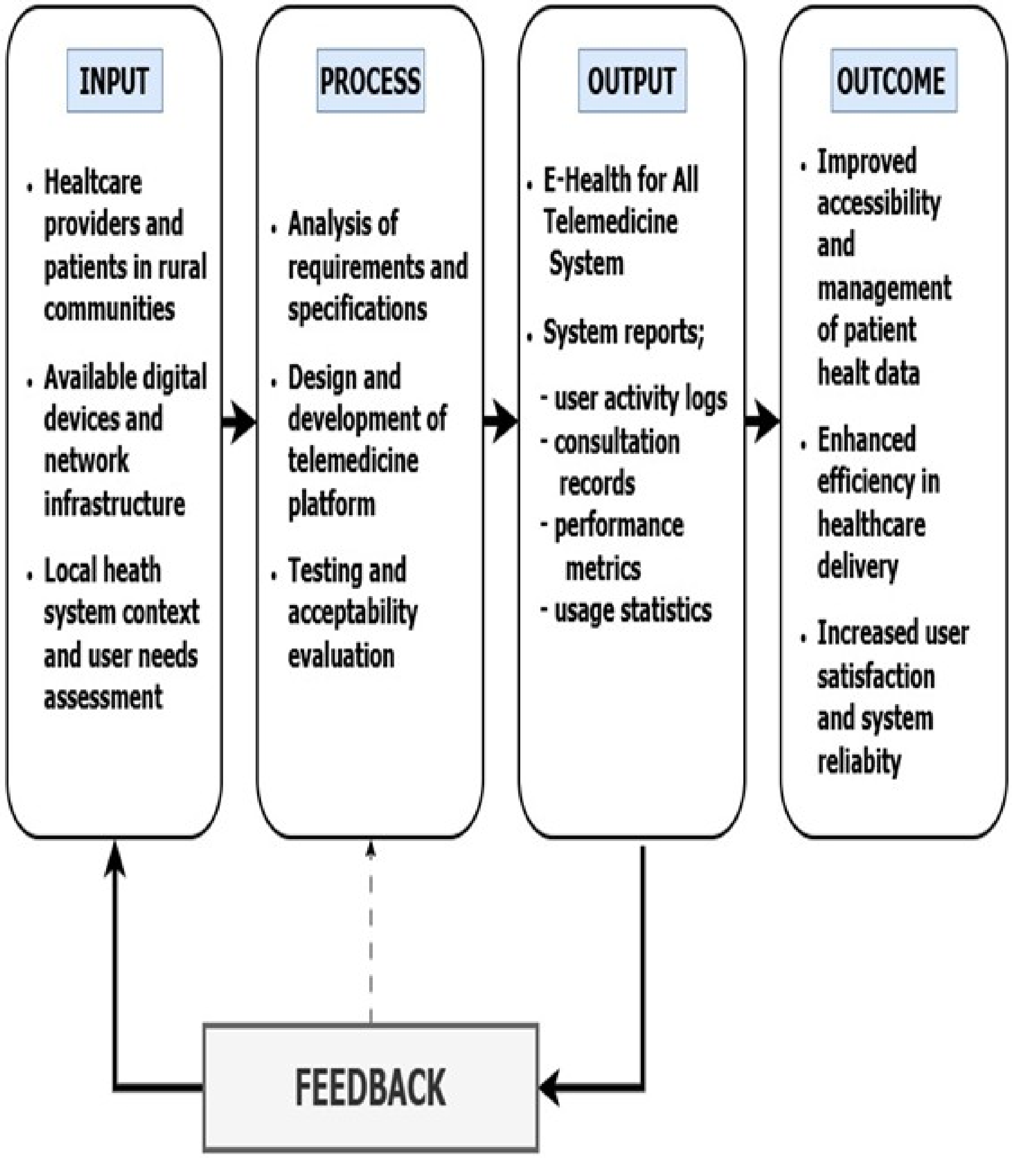

Figure 1.2. Conceptual Paradigm based on the Input-Process-Output-Outcome (IPOO) model illustrating the systematic development of the E-Health for All telemedicine system.

c) Data security

1.7 Assumptions

The research study is anchored on the following key assumptions:

1. Rural healthcare providers in Panicuason, Naga City have specific practices for managing patient records, scheduling doctors, and handling appointments.
2. There are challenges with the current systems for managing patient records, scheduling doctors, and setting up appointments in rural healthcare facilities.

3. A user-centered telemedicine platform can be designed and developed to tackle these challenges in the current healthcare delivery system.
4. The rural healthcare providers and clinic staff are ready to accept and shift from current manual practices toward the proposed user-friendly telemedicine platform and digital records management system.

1.8 Scope and Delimitation

This study addresses healthcare accessibility problems in rural populations, specifically focusing on rural healthcare providers and their patients. The objective of this study is to design a user-friendly online system to improve the management of patient records, aid in scheduling providers, and assist in arranging appointments to enable healthcare delivery to be more effective in rural environments.

The study will be conducted in Barangay Panicuason, Naga City, covering the entire design and development of a prototype for telemedicine functions, including digital patient health profiles, clinical encounter documentation, vital signs monitoring, medication monitoring, medical history access, and an appointment scheduling system supported by an integrated doctor database. The platform is a web application using a MySQL database and PHP, HTML, CSS, and JavaScript with XAMPP local server for testing. It works on Windows 10/11, macOS, Linux, and Android devices offline. The system only supports Android smartphones running version 8.0 or higher with a minimum screen resolution of 720×1280 pixels.

However, this study is limited only to Barangay Panicuason and does not extend to other rural or urban healthcare facilities, which narrows the generalizability of the findings. The system does not include high-level features such as lab integration, payment and billing modules, pharmacy management, patient portals, or iOS compatibility. Factors such as internet instability, power outages, varying digital literacy levels, and changes in healthcare policies are also beyond the study's scope.

1.9 Significance of the Study

This research is important for various stakeholders and contributes to improving healthcare equity and digital health innovation. The following are the key beneficiaries:

Patients in Rural Communities. Rural residents will see the most direct benefits by avoiding expensive trips to urban medical facilities for routine check-ups. Patients save money on transportation and time spent traveling, allowing them to access healthcare without disrupting work or family responsibilities. Digital medical records give patients more health information, while the appointment system cuts down on waiting times.

Rural Healthcare Providers. Healthcare providers in rural health units will benefit from automated patient record management and better appointment scheduling. Digital patient records allow for quicker clinical decisions, especially during emergencies. The integrated doctors' database makes it easier to refer patients to specialists and coordinate care.

Community Health Workers. Frontline workers gain better capabilities through quick access to patient information for consultations and emergencies. The system reduces paperwork, allowing them to focus on direct patient interaction and health education. Workers can help facilitate remote consultations for patients uncomfortable with technology and track health trends to spot emerging issues early.

Health Administrators and Local Government Officials. Administrators get valuable data on how telemedicine helps fill gaps in rural healthcare, supporting better resource allocation decisions. The platform's reporting features allow tracking of disease prevalence, service use, and community healthcare needs.

Technology Developers. Technology developers get practical insights into the technical needs and key features that are most important to rural healthcare stakeholders. The research outlines detailed requirements for building systems that can work offline and function well even with unreliable internet access.

Academic Researchers. The research community benefits from clear frameworks for creating easy-to-use digital health solutions for rural areas with limited resources. This study offers evidence on effective design methods and implementation techniques specifically made for resource-limited settings, including validated tools for measuring how acceptable telemedicine is.

1.10 Definition of Terms

E-health. E-health, also called electronic health or digital health, refers to using information and communication technologies in healthcare delivery [58]. This includes telemedicine, electronic medical records, and digital health platforms. It covers any health-related service or information provided electronically through the internet, mobile devices, and telecommunications systems to improve healthcare access, quality, and efficiency.

E-health for All. The main principle and title of this research. It highlights the need for equal access to digital health technologies for everyone, regardless of where they live, their income, or their technology skills [59]. In this study, it means committing to creating inclusive, user-friendly telemedicine solutions aimed specifically at rural communities in Naga City.

Telemedicine. The delivery of healthcare services and clinical information through communication technologies [60]. This approach helps overcome geographical barriers and improves access to medical services for patients in remote areas.

User-centered design. A development methodology that focuses on the needs, preferences, and limitations of end-users during the design process [61]. It involves actively engaging users in defining requirements, evaluating prototypes, and refining systems.

Rural healthcare. Providing medical services in isolated or underserved areas [62]. These locations often lack adequate healthcare infrastructure, have few medical specialists, and face significant barriers to quality health services.

Telehealth platform. A digital health ecosystem that combines various functions [63], including provider directories, patient information systems, and scheduling tools to facilitate remote healthcare delivery and improve care coordination.

Healthcare accessibility. How easily individuals can obtain necessary health services on time. It considers factors like physical availability, geographical proximity, economic affordability, and organizational fit.

CHAPTER 2: METHODS

2.1 Research Design

This study will adopt a developmental-descriptive qualitative research design. This approach was chosen to describe and understand the current healthcare access challenges in rural communities and to identify how telemedicine can offer solutions. The descriptive method will help document the current state of healthcare delivery and accessibility, and will provide detailed insights from those who would use the telemedicine system daily.

Data will be collected through interviews, observations, and surveys, providing understanding of how rural residents, healthcare providers, and community health workers experience the current healthcare access situation. The study will also include documentary analysis, where the researchers will review documents and observe how healthcare services are accessed in these communities.

The study has four main phases: Phase 1 is the *preparation phase*, where approvals are obtained and research tools are prepared. Phase 2 is the *assessment phase*, during which observation and documentation of the current healthcare access situation are conducted. Phase 3 is the *data collection phase*, where the researchers conduct interviews and gather feedback. Phase 4 is the *analysis phase*, where all information is organized to make recommendations.

2.1.1 Software Methodology

The software development methodology for the E-Health for All application will follow an iterative and incremental approach, combining elements of the Agile methodology. Agile operates in short cycles where developers create small parts, present them to users, gather feedback, and make improvements before proceeding.

Development Phases. The development will start with a list of all necessary features based on research. Developers will create simple mockups and share them with healthcare providers and rural residents. Programming will then begin in phases:

1. User registration and doctor database
2. Appointment scheduling
3. Video consultations and patient records
4. Prescription management

Testing Process. Testing will occur before launch. Developers will test the platform on different devices and internet speeds. Healthcare providers will conduct mock consultations, while rural residents will test the mobile app. Security testing will ensure that patient data is protected.

Pilot Implementation. Pilot testing in selected barangays will identify issues before the full rollout. Training will be provided to healthcare providers, community health workers, and rural residents.

2.2 System Design

The system design process employed several modeling techniques such as use case and system workflows to map user interactions and operational processes. These techniques directly captured the functional requirements, user actions, and step-by-step procedures within the system.

2.2.1 Use Case

The use case diagram shows the specific interactions between patients, healthcare providers, and system administrators with the system. In the use case diagram, there were three actors: patients, doctors, and system administrators. All actors can register and log in to the system. Patients and doctors utilize a mobile application, while system administrators access a dedicated website application.

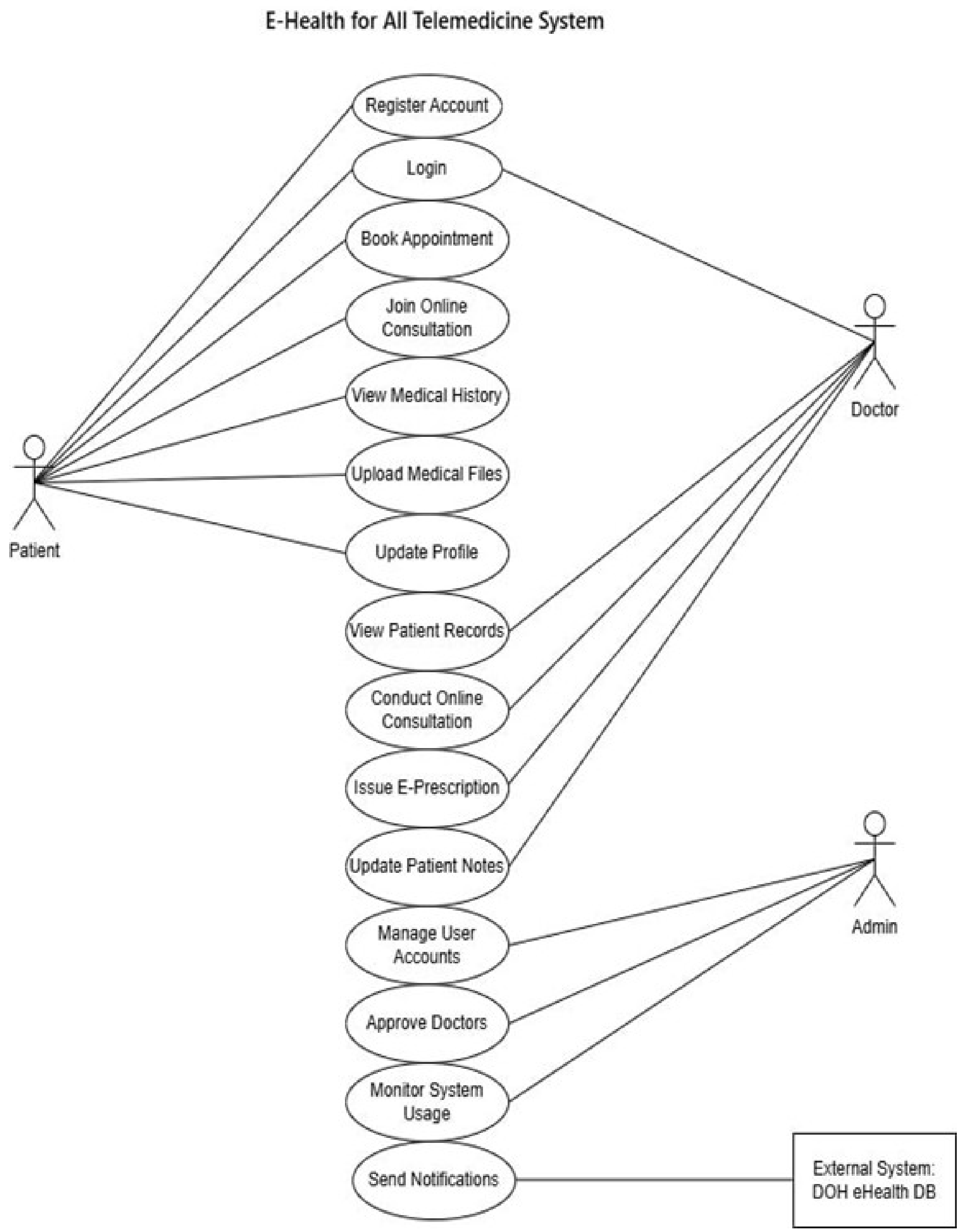

Figure 2.1. Use Case Diagram of the E-Health for All Telemedicine System.

2.2.2 System Workflows

The system workflows illustrate the sequential processes and decision points for key operations within the E-Health for All Telemedicine System. These workflows ensure standardized procedures and guide users through various system functionalities.

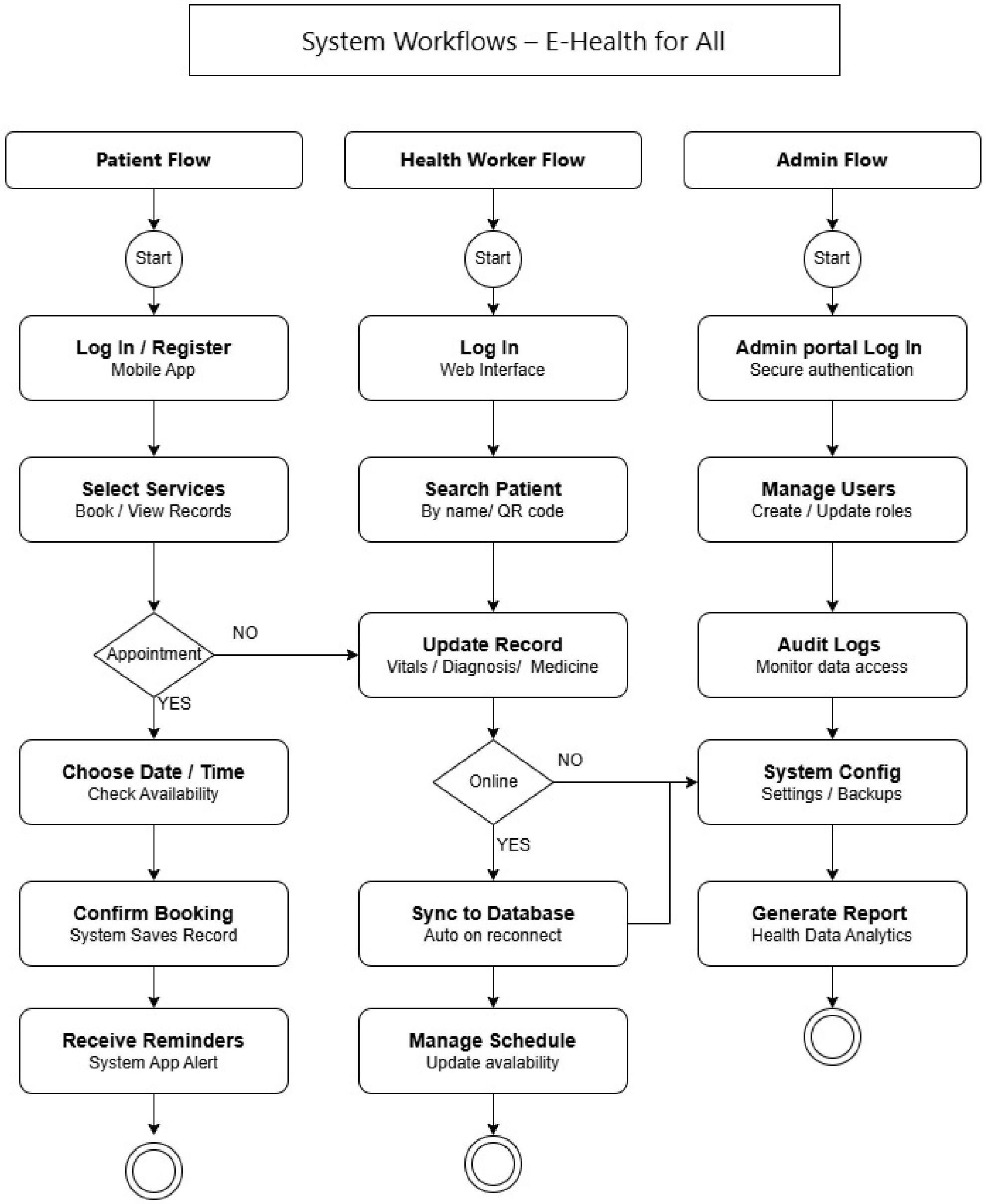

Figure 2.2. System Workflows of the E-Health for All Telemedicine System.

2.2.3 Systems Architecture

The system architecture for the E-Health for All Telemedicine System integrates three key stakeholders: Healthcare Providers (Doctors), Rural residents, and System Administrators. The proposed system comprises three main components reflected in the Application Layer of the architecture:

1. **Web-based platform for healthcare providers**—includes secure login, patient record management, video consultation features, and the ability to document

symptoms, diagnoses, and prescriptions.

2. **Mobile application for rural residents**—allows users to schedule appointments, access medical records, conduct video consultations, receive appointment reminders via SMS, and access health education information.
3. **Doctor database and scheduling system**—maintains provider profiles including specializations, qualifications, availability, and consultation fees.

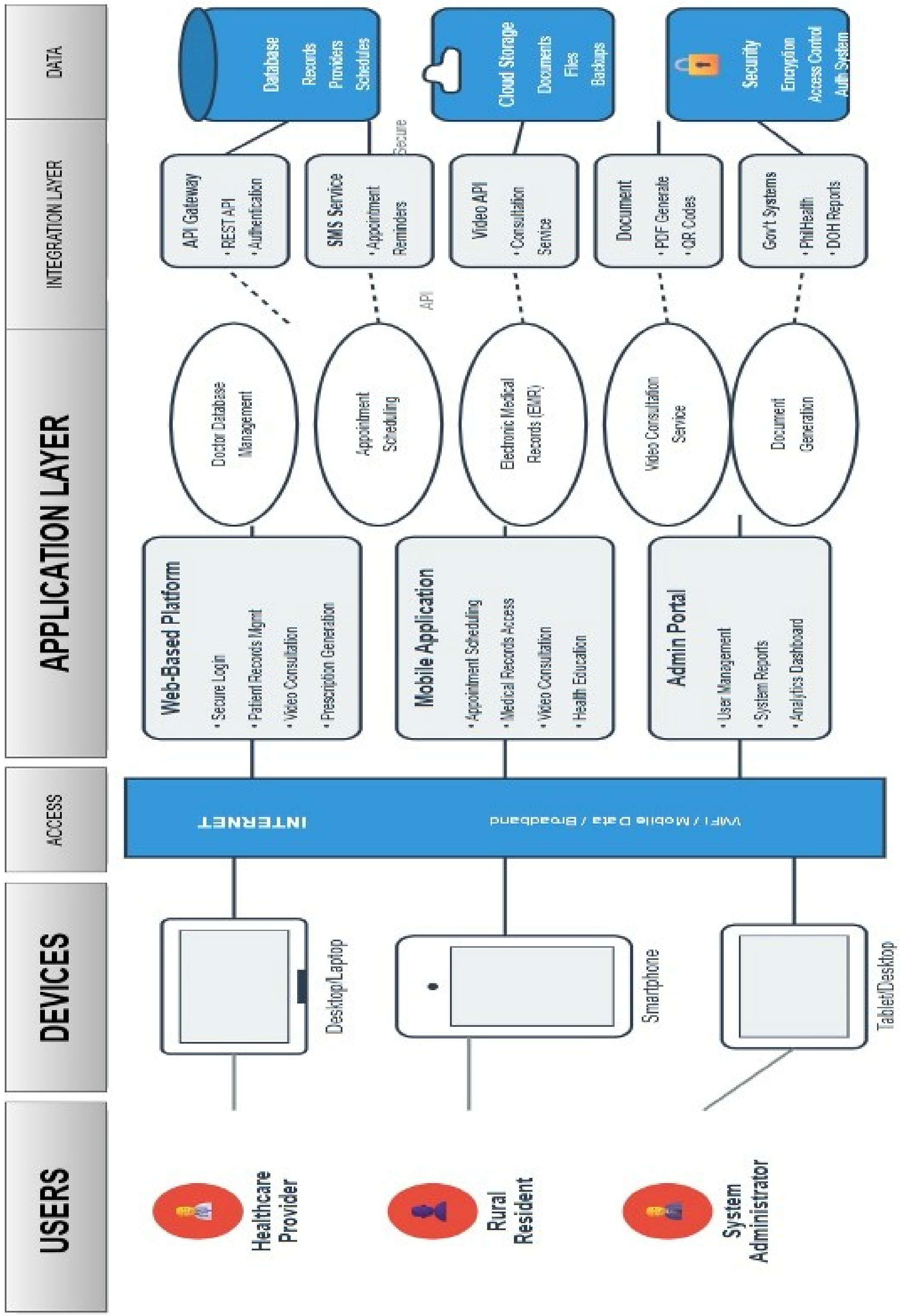

Figure 2.3. System Architecture of the E-Health for All Telemedicine System.

2.2.4 Context Diagram (DFD Level 0)

The Context Diagram shows the entire data flow of the E-Health for All Telemedicine System, where four external entities Patient, Doctor, BHW, and Admin exchange certain data with the system. This diagram serves as a foundation for designing the system. The data coming into and out of the system is clearly represented.

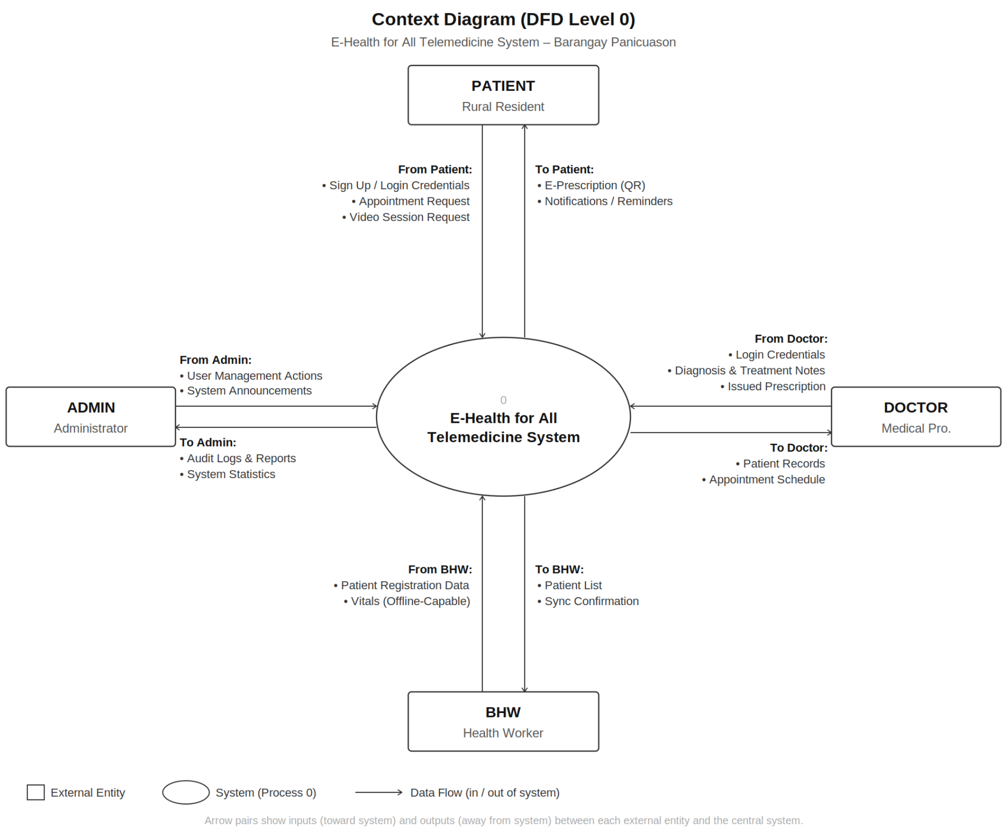

Figure 2.4. Context Diagram (DFD Level 0) of the E-Health for All Telemedicine System.

2.2.5 Data Flow Diagram Level 1

The Level 1 DFD illustrates the detailed internal processes and data flows within the Health Connect System. It expands the context diagram into eight (8) major processes supported by different data stores and external entities.

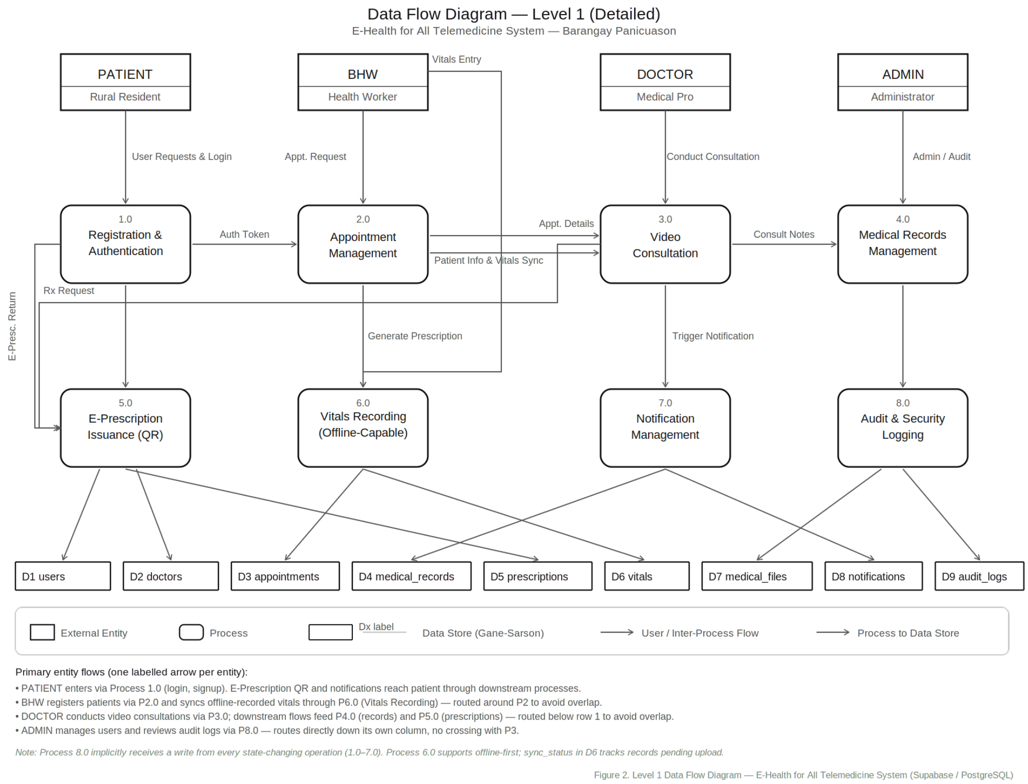

Figure 2.5. Data Flow Diagram — Level 1 (Detailed) of the E-Health for All Telemedicine System.

2.2.6 Entity Relationship Diagram

The ERD visualizes each entity within the database design of the system, which is characterized by nine (9) entities, namely Users, Doctors, Appointments, Medical Records, Prescriptions, Vitals, Notifications, Audit Logs, and Medical Files. It further shows how these entities are linked through their relationships and foreign keys, thus representing the logical arrangement of the database within the system.

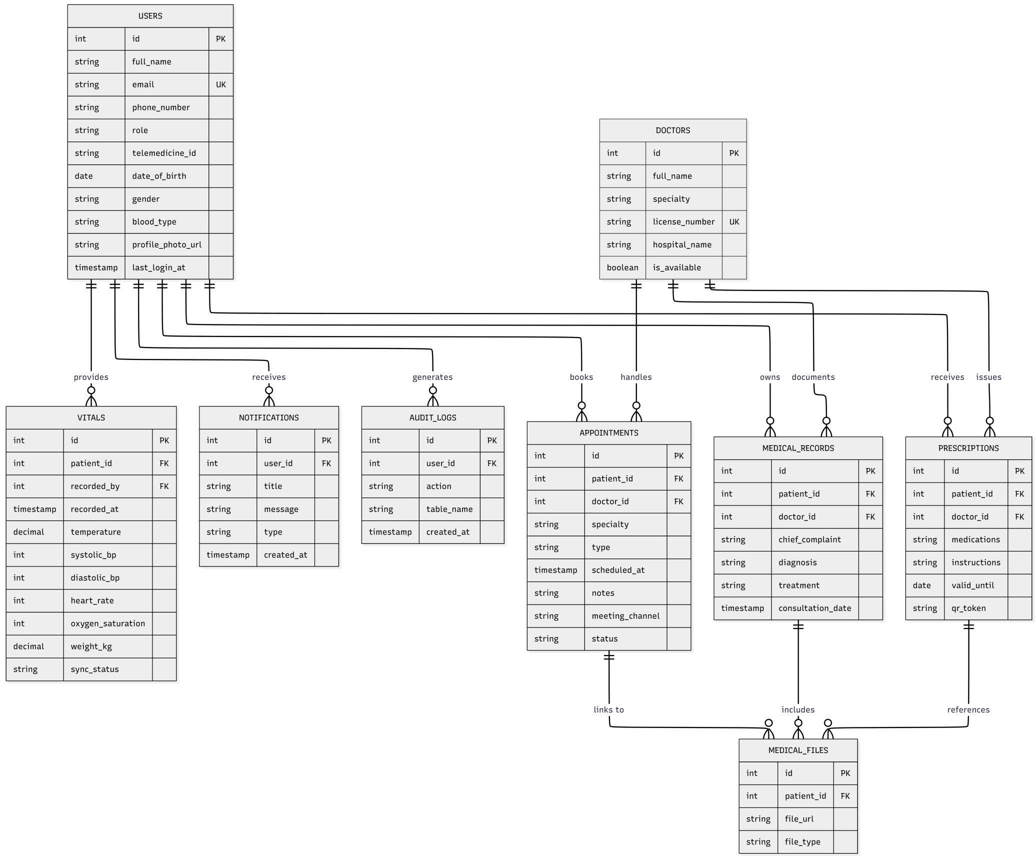

Figure 2.6. Entity Relationship Diagram of the E-Health for All Telemedicine System.

2.3 Population and Sample

The researcher determined the respondents through a systematic selection process based on their direct involvement with rural healthcare delivery at Panicuason Barangay Health Center in Naga City. The study involved healthcare providers and staff from Panicuason Rural Health Unit in Naga City.

Based on preliminary observation, the barangay health center has a total of 11 healthcare personnel consisting of 7 Dental Health Workers (DHW), 1 Midwife, 1 Encoder, 1 Barangay Nutrition Scholar (BNS), and 1 Staff Nurse (SNF). Since there are only 11 respondents, the study will utilize total enumeration as the sampling procedure for the design of the system.

For the rural community residents, purposive sampling will be implemented. The sample size for rural residents is determined using Slovin's formula:

$$n = \frac{N}{1 + Ne^2} \tag{2.1}$$

where:

- n = required sample size
- N = total population of Barangay Panicuason
- e = margin of error (set at 0.05 or 5%)

This formula helps ensure our sample size is adequate to represent the barangay population while maintaining statistical reliability.

2.4 Research Locale

The study will be conducted at Barangay Panicuason Rural Health Unit in Naga City, Camarines Sur. Barangay Panicuason is located outside the urban center of Naga City and faces typical rural healthcare challenges including limited specialist availability and equipment, long travel distances to medical facilities, and insufficient healthcare infrastructure.

The Rural Health Unit serves as the primary healthcare facility for residents of Barangay Panicuason and surrounding areas. The facility provides basic health services including primary care consultations, maternal and child health programs, immunization services, disease prevention activities, and health education.

2.5 Data Gathering Procedure

Preparation Phase. This is the first step in establishing formal agreements and obtaining necessary approvals before starting the research. It involves getting approval and preparing formal letters to request permission from barangay officials and local health officers, and creating consent forms for all participants.

Document Analysis. This method collects data by examining health records from barangay health stations, existing referral systems, and community health profiles. The researchers will analyze: types of health complaints commonly addressed, frequency of referrals to distant facilities, travel time and costs to access healthcare, availability of medical specialists, and community health statistics.

Direct Observation. This method involves systematically watching and recording healthcare access patterns and challenges in their natural setting. The researchers will take detailed notes about healthcare-seeking behavior, time travel to the nearest clinic or hospital, document available transportation options, and observe how health workers currently handle patient consultations and referrals.

Surveys. This method collects information from respondents through standardized questions and response options. It gives respondents a chance to share their views, preferences, and experiences about current healthcare access and their needs for a telemedicine solution.

Interviews. This method gathers information through one-on-one conversations with stakeholders, allowing for deeper exploration of issues and concerns. The researchers will conduct individual interviews lasting 30 to 45 minutes with rural residents, healthcare providers, and IT personnel.

2.6 Validation and Testing Methods

Functionality Testing. Unit testing verified individual components, with each function tested independently to ensure correct performance. Test cases covered normal inputs, edge cases, and invalid inputs. Integration testing verified module interactions, ensuring data flowed correctly between modules.

User Acceptance Testing. Healthcare providers evaluated whether the system met their needs. Sign-off criteria required at least 80 percent feature acceptance. Features falling below this threshold were revised based on feedback.

Performance Testing. Load testing simulated multiple concurrent users, testing the system with 10, 20, and 30 simultaneous users to ensure acceptable performance under typical and peak loads. Target response times were set under 3 seconds for most operations.

Security Testing. Access control verification ensured users could only access features appropriate to their roles. Data encryption validation confirmed sensitive information was encrypted in transit using HTTPS and at rest in the database.

Usability Testing. The think-aloud protocol was employed with 5 to 8 representative users. Task completion rates measured how many users successfully completed tasks. Time-on-task measured efficiency. Error rates counted mistakes made during tasks.

2.7 Research Instrument

Interview Guides. Researchers will prepare interview guides for rural residents of Barangay Panicuason covering: experiences with healthcare services at the barangay health center, distance traveled to see a doctor, costs involved in getting medical care, health problems commonly faced, access to mobile phones and the internet, comfort level with technology, and desired features in a telemedicine platform.

Survey Questionnaires. For rural residents, the questionnaire will ask about their age, household size, and common health conditions. Other questions will include: satisfaction with current healthcare access using a 5-point scale, distance traveled to see a doctor, healthcare spending, frequency of delaying medical care, mobile phone and internet access, and preferred telemedicine features.

2.8 Ethical Considerations

Voluntary Participation. Participation in this study is completely voluntary. No one will be pressured or given incentives that could unduly influence their decision to join. Participants can withdraw at any time without consequences.

Informed Consent. All participants received detailed information about the study before agreeing to participate. This information covered the study's purpose, what participation involved, how data would be collected and used, potential risks and benefits, confidentiality measures, and the right to withdraw.

Parental/Guardian Consent and Minor Assent. For participants under 18 years of age, written parental or legal guardian consent will be obtained before any research involvement. Minors aged 7–17 years will also provide written assent. Minors may decline participation even if parents have provided consent.

Privacy and Confidentiality. Individual identities will not be revealed without explicit written permission. Interview recordings and transcripts will be stored in password-protected digital files accessible only to the research team.

Data Protection. All research data will be handled according to the Data Privacy Act of 2012 (Republic Act No. 10173). Personal identifiers will be removed from datasets, and survey responses will remain anonymous.

2.9 Statistical Tools

2.9.1 Descriptive Statistics

The study employed descriptive statistical methods to summarize and describe the characteristics of the sample and their responses.

Percentage. To show the proportion of respondents who chose each answer option, percentages are calculated using the formula:

$$P = \frac{f}{N} \times 100 \tag{2.2}$$

where f is the frequency or number of responses for a specific option and N is the total number of respondents.

Weighted Mean. For Likert-scale items, the weighted mean is calculated using the formula:

$$\bar{x} = \frac{\sum(W \cdot f)}{N} \tag{2.3}$$

where W is the weight or scale value (1–5 for a 5-point Likert scale), f is the frequency of respondents who chose that value, and N is the total number of respondents.

2.9.2 Frequency Distribution

Frequency tables display how many respondents selected each response option. Table 2.1 shows a sample frequency distribution for a 5-point satisfaction scale.

Table 2.1. Sample Frequency Distribution of Respondent Satisfaction

Response Category	No. of Respondents	Percentage
Very Satisfied	5	8.3%
Satisfied	25	41.7%
Neutral	15	25.0%
Dissatisfied	10	16.7%
Very Dissatisfied	5	8.3%
Total	**60**	**100%**

2.9.3 Statistical Treatment of Data

The data gathered will be interpreted using a five-point Likert scale to measure the respondents' attitudes, opinions, perceptions, and preferences regarding the user-centered telemedicine system. Table 2.2 shows the interpretation scale used for weighted mean values.

Table 2.2. Likert Scale for Agreement on Telemedicine System Acceptability

Scale	Range	Interpretation
5	4.21 – 5.00	Strongly Agree
4	3.41 – 4.20	Agree
3	2.61 – 3.40	Neutral
2	1.81 – 2.60	Disagree
1	1.00 – 1.80	Strongly Disagree

2.10 Project Timeline

Table 2.3 outlines the systematic schedule of activities for conducting the research study on the user-centered telemedicine system.

Table 2.3. Project Timeline from Preparation Phase to Prototype Implementation

Activity	M1	M2	M3	M4	M5	M6	M7	M8
Preparation Phase								
Finalize proposal and instruments	●	●						
Obtain approvals and permissions	●	●						
Pilot test instruments	○	○						
Data Collection – Observations								
Observation in Panicuason		●	●					
Document analysis		●	●					
Data Collection – Stakeholder Input								
Individual interviews			●	●				
Survey distribution and collection			●	●				
Data Analysis								
Transcription					●	●		
Coding and thematic analysis					○	●		
Statistical analysis						○		
Report Writing								
Findings documentation						○	○	●
Final report preparation							●	●

Legend: ● = Active work period; ○ = Overlap or transitional period

CHAPTER 3: RESULTS AND DISCUSSIONS

This chapter presents the findings gathered from the survey administered to the health facility personnel of Barangay Panicuason Rural Health Unit in Naga City. Results are organized according to the specific research objectives: (1) current healthcare practices, (2) challenges encountered in the current system, (3) system design plans as the prototype response to identified challenges, and (4) the level of acceptability of the initial prototype in terms of usability, reliability, and data security. Data are presented using weighted mean scores and frequency distributions, discussed in relation to the reviewed literature, and linked to the theoretical frameworks guiding the study.

3.1 Profile of Respondents

A total of 13 health facility personnel completed the survey questionnaire during the data-gathering phase. Table 3.1 presents the demographic profile of the respondents.

The respondents were predominantly female (92.3%), consistent with the largely female composition of community health workforces in Panicuason. The majority were aged 25–34 (46.2%), indicating a relatively young and potentially technology-receptive group. In terms of work experience, most respondents had 10 years or more of service (61.54%), followed by those with 4–6 years (30.77%), while only a small proportion had 1–3 years (7.69%) and none fell within the 7–9 year range. This suggests that the respondents generally possess substantial professional experience, enabling them to provide informed and reliable assessments of current operational practices. These demographics align with the Task–Technology Fit (TTF) theory's premise that technology adoption is influenced by users' experiential familiarity with their work tasks [56].

3.2 Current Practices of Rural Healthcare Providers

Research Question 1 asked: "What are the current practices of rural healthcare providers with respect to patient records management, doctor and health worker scheduling, and appointment systems?" Table 3.2 presents the weighted mean scores for each indicator. *Scale: 5 = Always (4.21–5.00), 4 = Often (3.41–4.20), 3 = Sometimes (2.61–3.40), 2 = Rarely (1.81–2.60), 1 = Never (1.00–1.80)*

The results revealed consistently high practice scores across all three domains. Patient records management registered the highest overall weighted mean of 4.60 (Always), indicating that health facility personnel demonstrate strong commitment to documenting, organizing, and retrieving patient records. Notably, records are documented immediately

Table 3.1. Demographic Profile of Survey Respondents (n = 13)

Category	Frequency (f)	Percentage (%)
A. Sex		
Female	12	92.3%
Male	1	7.7%
Total	13	100%
B. Position / Designation		
Dental Health Worker (DHW)	7	53.8%
Midwife	1	7.7%
Encoder	1	7.7%
Barangay Nutrition Scholar (BNS)	1	7.7%
Community Based Rehabilitation Specialist (CBRS)	1	7.7%
Barangay Social Welfare Aide (BSWA)	1	7.7%
Total	13	100%
C. Age Bracket		
Below 25 years old	1	7.7%
25–34 years old	6	46.2%
35–44 years old	4	30.8%
45 years old and above	2	15.4%
Total	13	100%
D. Years of Experience		
Less than 1 year	0	0%
1–3 years	1	7.7%
4–6 years	3	30.77%
7–9 years	0	0%
10 years and above	8	61.54%
Total	13	100%

after consultations (WM = 4.70), reflecting adherence to standard clinical documentation protocols.

Doctor and health worker scheduling obtained an overall WM of 4.28 (Often), with backup arrangement scoring highest at 4.50 (Always), suggesting that contingency planning is well-established at the facility. Appointment systems averaged 4.38 (Often), with systematic arrangement and advance patient notification both scoring 4.50 (Always). These high practice scores align with findings by Santos, Mendoza, and Cruz [4], who noted that Philippine rural health units have established basic operational protocols, though implementation quality varies significantly between facilities.

These findings support the Task–Technology Fit theory [56] in that the existing manual practices are well-established and purposive. A digital system that replicates and enhances these existing workflows rather than replacing them is more likely to achieve strong

Table 3.2. Current Healthcare Practices — Weighted Mean Scores (n = 13)

Indicator	WM	Interpretation
A. Patient Records Management		
2.1 Records documented immediately after each consultation	4.70	Always
2.2 Records stored in an organized and accessible manner	4.60	Always
2.3 Records updated regularly to reflect current health status	4.50	Always
2.4 Records retrieved quickly and accurately when needed	4.60	Always
Overall Weighted Mean — Patient Records Management	**4.60**	**Always**
B. Doctor and Health Worker Scheduling		
2.5 Schedules planned and communicated in advance	4.20	Often
2.6 Availability clearly recorded and accessible to staff	4.20	Often
2.7 Schedule changes updated and communicated promptly	4.20	Often
2.8 Backup/replacement arranged when scheduled worker is unavailable	4.50	Always
Overall Weighted Mean — Doctor Scheduling	**4.28**	**Often**
C. Appointment Systems		
2.9 Appointments arranged in an orderly and systematic manner	4.50	Always
2.10 Patients informed of appointment schedules in advance	4.50	Always
2.11 Appointment records maintained accurately to avoid double-booking	4.30	Often
2.12 Patients attended within a reasonable waiting time	4.20	Often
Overall Weighted Mean — Appointment Systems	**4.38**	**Often**

adoption among health workers who are already familiar with structured documentation and scheduling routines [64].

3.3 Challenges Encountered in the Current System

Research Question 2 asked: "What are the challenges encountered in the current system?" Table 3.3 presents the weighted mean scores for each challenge indicator.

Scale: 5 = Always (4.21–5.00), 4 = Often (3.41–4.20), 3 = Sometimes (2.61–3.40), 2 = Rarely (1.81–2.60), 1 = Never (1.00–1.80)

All challenge indicators were rated in the "Sometimes" range (WM = 2.61–3.40), confirming that while problems were not constant, they occurred with enough regularity to warrant systematic intervention. The most frequently cited challenge across domains was the difficulty in organizing and managing patient records (WM = 3.20), consistent with

Table 3.3. Challenges in Current Healthcare Practices — Weighted Mean Scores (n = 13)

Challenge Indicator	WM	Interpretation
A. Patient Records Management		
3.1 Patient records are difficult to organize and manage	3.20	Sometimes
3.2 Patient records are missing or incomplete	2.80	Sometimes
3.3 Retrieving a patient's record takes too much time	3.00	Sometimes
3.4 Records are lost or damaged due to current storage method	2.80	Sometimes
Overall Weighted Mean — Patient Records Challenges	**2.95**	**Sometimes**
B. Doctor and Health Worker Scheduling		
3.5 Scheduling conflicts among doctors/health workers occur	3.00	Sometimes
3.6 No clear or updated record of availability	2.90	Sometimes
3.7 Health workers assigned to multiple duties simultaneously	2.70	Sometimes
3.8 Schedule changes not communicated to concerned staff	2.70	Sometimes
Overall Weighted Mean — Scheduling Challenges	**2.83**	**Sometimes**
C. Appointment Systems		
3.9 Patients experience long waiting times before being attended to	3.00	Sometimes
3.10 Delays in processing patient appointments	2.90	Sometimes
3.11 Double-bookings or missed appointments occur	2.70	Sometimes
3.12 No organized system for tracking or confirming appointments	3.00	Sometimes
Overall Weighted Mean — Appointment Challenges	**2.90**	**Sometimes**

findings by D'Amore et al. [17] that paper-based record systems in rural health facilities are particularly susceptible to organizational lapses due to high patient volumes and limited administrative staffing.

Scheduling challenges averaged 2.83, with scheduling conflicts (WM = 3.00) and unclear availability records (WM = 2.90) as primary concerns. This aligns with Mendoza, Cruz, and Santos [44], who found that the absence of a centralized scheduling system in Philippine rural health units contributes to coordination failures, particularly during peak consultation periods. Appointment system challenges averaged 2.90, with the absence of a tracking system (WM = 3.00) and long patient waiting times (WM = 3.00) as the most persistent issues. Del Rosario [52] similarly found that the lack of digital appointment management in Philippine rural health centers contributes significantly to extended waiting times and patient dissatisfaction.

These "Sometimes" ratings, while not alarming individually, collectively defined a pattern of recurring inefficiency that compounds over time. They validated the need for a digital solution that addresses record organization, scheduling conflict detection, appointment tracking, and automated reminders—all of which are core features of the proposed E-Health for All system [65].

3.4 System Design and Plans for the Prototype

Research Question 3 asked: "What system can be designed to address the identified challenges?" This section presents the importance ratings for proposed system features. Table 3.4 describes the design plans for the prototype, including the use case, workflow, and database schema.

Scale: 5 = Very Important (4.21–5.00), 4 = Important (3.41–4.20), 3 = Moderately Important (2.61–3.40), 2 = Slightly Important (1.81–2.60), 1 = Not Important (1.00–1.80)

Table 3.4. Importance of Proposed Features — Weighted Mean Scores (n = 13)

Proposed System Feature	WM	Interpretation
4.1 Digital storage and easy retrieval of patient records	4.50	Very Important
4.2 Automated doctor/worker scheduling with conflict detection	4.50	Very Important
4.3 Online patient appointment booking	4.60	Very Important
4.4 Automated appointment reminders and notifications	4.80	Very Important
4.5 Secure login and role-based access control	4.90	Very Important
Overall Weighted Mean	**4.66**	**Very Important**

All five proposed features received "Very Important" ratings, with an overall weighted mean of 4.66. Secure login and role-based access control (WM = 4.90) ranked highest, reflecting a strong awareness among health workers of data privacy obligations under the Data Privacy Act of 2012. Automated reminders (WM = 4.80) ranked second, directly addressing the challenge of missed appointments (Item 3.11, WM = 2.70) and long waiting times (Item 3.9, WM = 3.00). Online appointment booking (WM = 4.60) and digital storage (WM = 4.50) also scored highly, validating the core functional design of the proposed system. These ratings are consistent with the TAM framework [57], where perceived usefulness—as evidenced by the high importance ratings—is a primary determinant of technology adoption intention.

3.5 Summary of Findings

This section provides a concise synthesis of the study's key findings and reflects on their broader significance in advancing rural healthcare delivery through digital innovation.

This study aimed to design and develop E-Health for All, a user-centered telemedicine platform for Barangay Panicuason Rural Health Unit in Naga City, addressing inefficiencies in patient records management, health worker scheduling, and appointment systems. Using a developmental-descriptive research design, data were gathered from 13 health facility personnel to document current practices, identify challenges, evaluate proposed system features, and assess prototype acceptability.

The findings reveal that health facility personnel had already established strong baseline practices across all three operational domains, indicating a structured and disciplined workforce. This is significant because it confirms that the E-Health for All system enters a context where existing workflows are well-established, making adoption more likely when the platform is designed to enhance rather than replace familiar routines—consistent with the Task–Technology Fit (TTF) theory.

Despite these strong practices, recurring challenges were confirmed across all domains. Difficulties in organizing physical records, the absence of a systematic appointment tracking mechanism, and scheduling conflicts among health workers were the most consistently reported concerns. These findings collectively validate the study's core rationale—that an integrated telemedicine platform is both necessary and timely for a facility of this scale and context.

All proposed system features were rated very important, with data security, automated reminders, and online booking receiving the highest endorsements—directly corresponding to the identified challenges. This alignment confirms that the system is grounded in users' actual needs, consistent with the Technology Acceptance Model (TAM). The respondent profile, predominantly female, relatively young, and experienced, further supports readiness for a managed digital transition as affirmed by the Diffusion of Innovations (DOI) theory [66].

3.6 Conclusions

Overall, this study demonstrates that E-Health for All is both technically feasible and operationally justified for Barangay Panicuason RHU. Beyond its immediate application, it offers a replicable framework for designing context-sensitive telemedicine solutions in similarly resource-limited settings.

3.7 Recommendations

Based on the conclusions of this study, the following recommendations are directed to the key stakeholders involved in the deployment, improvement, and continued research of the E-Health for All telemedicine system.

For the Barangay Panicuason Health Unit and the Local Government Unit. The local government unit should pursue the formal adoption and full deployment of the

E-Health for All system by allocating a dedicated budget for device procurement, internet connectivity, and system maintenance. A structured onboarding program should be conducted prior to deployment to ensure a smooth transition from paper-based to digital operations.

For System Developers and Future Development Teams. Developers should expand the system in future iterations to include laboratory result integration, pharmacy management, a patient-facing portal, and iOS compatibility. Offline functionality should be strengthened to maintain data access during internet outages, and all development must strictly uphold the Data Privacy Act of 2012 through encryption, role-based access controls, and regular security auditing.

For Health Facility Personnel. Health workers should actively engage during the pilot phase by documenting usability issues and communicating improvement suggestions through formal feedback channels. Their sustained adoption and cooperation will be essential to the system's continuous refinement and long-term success.

For Academic Researchers and Future Studies. Future researchers should extend this study by including community residents as secondary respondents and conducting longitudinal evaluations to assess long-term system impact. Comparative studies across multiple rural health units are recommended, and future instruments should incorporate the System Usability Scale (SUS) and Cronbach's Alpha to strengthen methodological rigor.

For Policymakers and the Department of Health. The Department of Health should consider the E-Health for All model as a pilot reference for rural health unit digitalization nationwide. Policymakers should provide technical and financial support to local government units pursuing similar initiatives and establish interoperability standards to connect locally developed platforms with national health systems such as iHOMIS and the National Health Data Repository.

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CURRICULUM VITAE

APPENDICES